Database Systems Lab

CSE 212

DBMS PROJECT REPORT

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SECTION : B1

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- Project Management System -

Description of Topic:

This system is designed to manage various aspects of projects, including tasks, teams, meetings, schedules, and reports. It involves several entities that interact with one another, allowing for efficient tracking of time, task assignments, notifications, reminders, and reporting. Key features include:

1. **User and Team Management:** Users are assigned to teams, which handle tasks and projects collaboratively.
2. **Task and Project Tracking**: Tasks are linked to projects with deadlines, statuses, and user assignments.
3. **Meeting Scheduling**: Meetings are organized to track progress or discuss tasks, with schedules specifying duration and time.
4. **Time Tracking:** Users can track the time spent on specific tasks for productivity measurement.
5. **Notifications and Reminders:** Reminders ensure deadlines are met, and notifications keep stakeholders informed.
6. **Client Interaction:** Projects can be linked to clients, with reports generated for progress updates.
7. **Comprehensive Reporting:** Reports summarize team and project performance, enabling better decision-making. This system provides an integrated approach to managing tasks, resources, and deadlines, ensuring collaboration and efficiency in project execution.

Entities from the schema:

1. User
2. Team
3. Client
4. Projects
5. Tasks
6. Meeting
7. Schedules
8. Reminder
9. Track_Time
10. Report

Attributes with Primary Key & Foreign Key:

1) **User :-**

- Attributes:
  - U_ID (Primary Key)
  - U_NAME
  - U_MAIL
  - TEAM_ID (Foreign Key, references TEAM (TEAM_ID))

2) **Team :-**

- Attributes:
  - TEAM_ID (Primary Key)
  - TEAM_NAME

3) **Client :-**

- Attributes:
  - C_ID (Primary Key)
  - C_NAME
  - C_MAIL

4) **Projects :-**

- Attributes:
  - P_ID (Primary Key)
  - P_NAME
  - START_DATE
  - END_DATE
  - TEAM_ID (Foreign Key, references Team(TEAM_ID))
  - C_ID (Foreign Key, references Client(C_ID))

5) **Tasks :-**

- Attributes:
  - T_ID (Primary Key)
  - T_NAME
  - T_DEADLINE
  - STATUS

    - U_ID (Foreign Key, references User(U_ID))
    - P_ID (Foreign Key, references Projects(P_ID))

6) **Meeting :-**

- Attributes:
    - M_ID (Primary Key)
    - M_TITLE
    - M_DATE_TIME
    - T_ID (Foreign Key, references Tasks(T_ID))
    - TEAM_ID (Foreign Key, references Team(TEAM_ID))

7) **Schedules :-**

- Attributes:
    - S_DATE_TIME
    - S_DURATION
    - M_ID (Foreign Key, references Meeting(M_ID))

8) **Reminder :-**

- Attributes:
    - R_ID (Primary Key)
    - R_TITLE
    - R_DATE_TIME
    - R_TYPE
    - M_ID (Foreign Key, references Meeting(M_ID))
    - T_ID (Foreign Key, references Tasks(T_ID))
    - U_ID (Foreign Key, references User(U_ID))

9) **Track_Time :-**

- Attributes:
    - TRACK_ID (Primary Key)
    - DATE_TIME
    - U_ID (Foreign Key, references User(U_ID))
    - T_ID (Foreign Key, references Tasks(T_ID))

10) **Report :-**

- Attributes:
    - REPORT_ID (Primary Key)
    - DATE
    - P_ID (Foreign Key, references Projects(P_ID))
    - TEAM_ID (Foreign Key, references Team(TEAM_ID))

Entities and Relationships:

One-to-Many Relationships:

1. **USER → TEAM**
    - A user can be associated with multiple teams. This is indicated by the foreign key U_ID in the **TEAM** table referencing USER.

2. **USER → TASKS**
    - A user can be assigned multiple tasks. This is shown by the foreign key U_ID in the **TASKS** table referencing USER.

3. **TEAM → MEETING**
    - A team can have multiple meetings. The **MEETING** table has a foreign key TEAM_ID referencing **TEAM**.

4. **PROJECTS → TASKS**
    - A project can have multiple tasks. This is shown by the foreign key P_ID in the **TASKS** table referencing **PROJECTS**.

5. **CLIENT → PROJECTS**
    - A client can have multiple projects. The **PROJECTS** table has a foreign key C_ID referencing **CLIENT**.

6. **TASKS → REMINDER**
    - A task can have multiple reminders. The **REMINDER** table has a foreign key T_ID referencing **TASKS**.

7. **MEETING → SCHEDULES**
    - A meeting can have multiple schedules. The **SCHEDULES** table has a foreign key M_ID referencing **MEETING**.

8. **USER → REMINDER**
    - A user can have multiple reminders. This is shown by the foreign key U_ID in the **REMINDER** table referencing **USER**.

9. **PROJECTS → REPORT**
    - A project can have multiple reports. The **REPORT** table has a foreign key P_ID referencing **PROJECTS**.

10. **TEAM → REPORT**
    - A team can have multiple reports. The **REPORT** table has a foreign key TEAM_ID referencing **TEAM**.

Many-to-One Relationships:

1. **TASKS → USER**
    - Many tasks can be assigned to a single user. This is shown by the foreign key U_ID in the **TASKS** table.

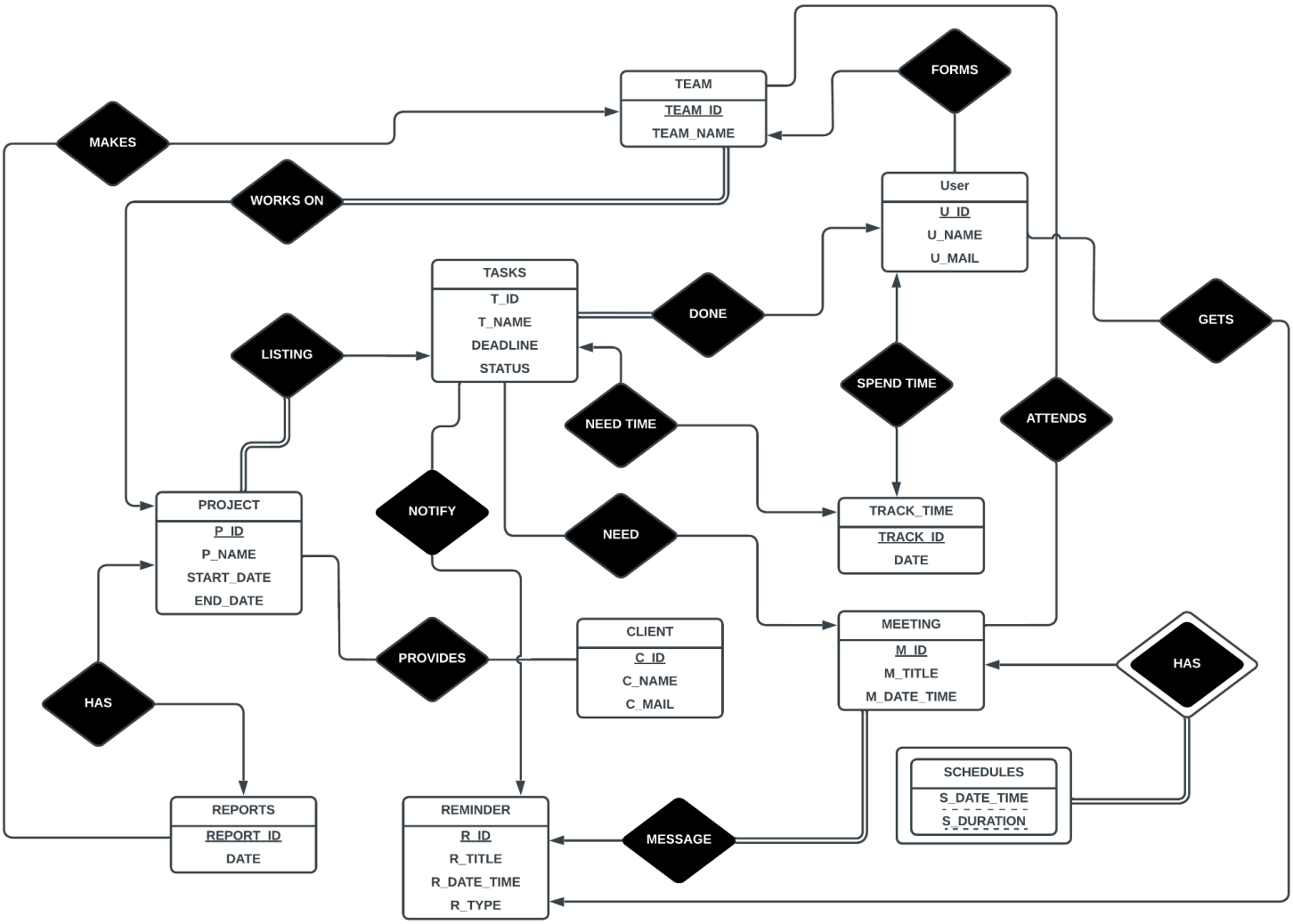
2. **PROJECTS → CLIENT**
    - Many projects can be associated with a single client. This is shown by the foreign key C_ID in the **PROJECTS** table.

3. **MEETING → TASKS**
    - Many meetings can be related to a single task. This is indicated by the foreign key T_ID in the **MEETING** table.

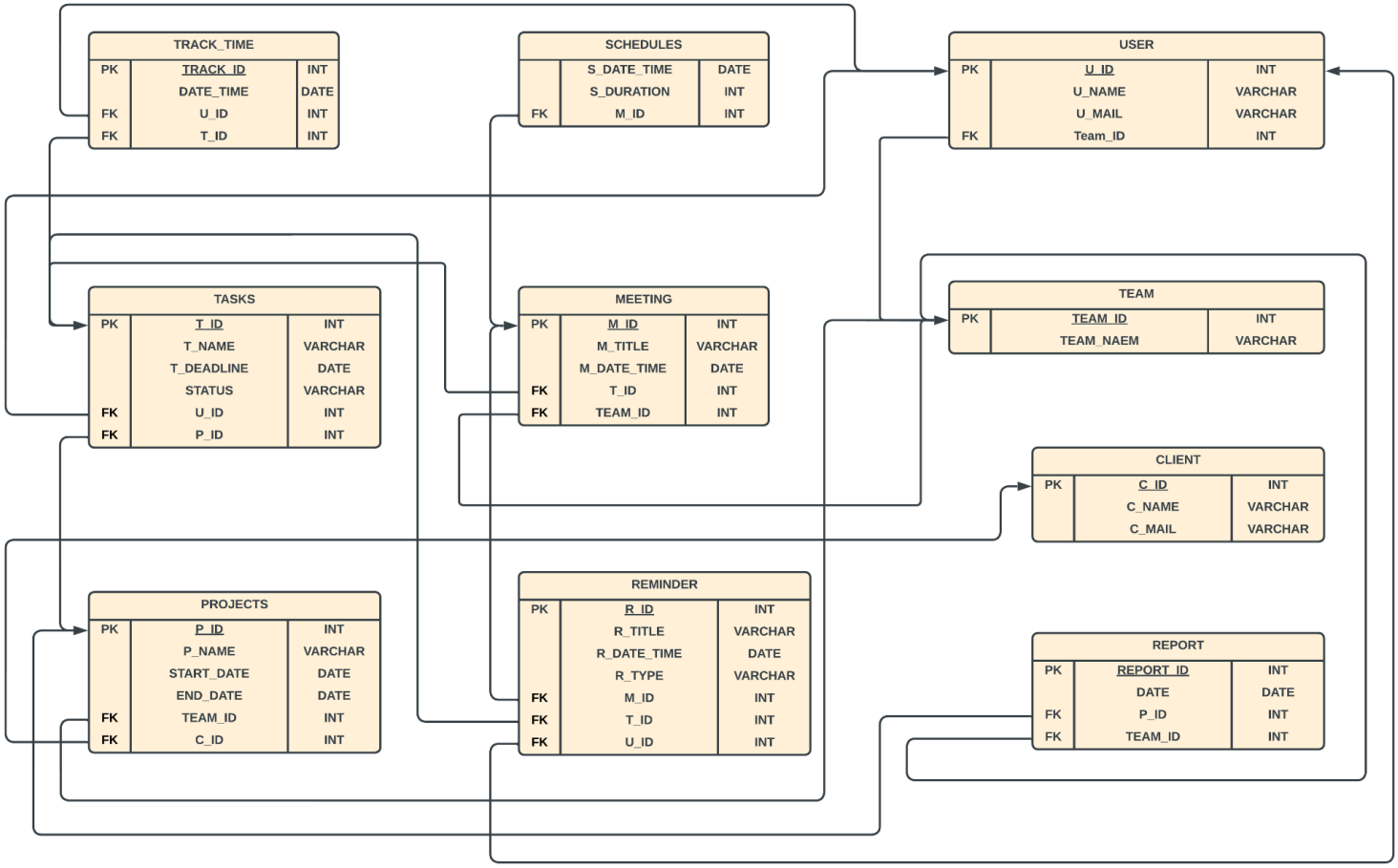
4. **REMINDER → TASKS**
    - Many reminders can be related to a single task. This is indicated by the foreign key T_ID in the **REMINDER** table.

5. **REMINDER → USER**
    - Many reminders can be assigned to a single user. This is shown by the foreign key U_ID in the **REMINDER** table.
    -
6. **REPORT → TEAM**
    - Many reports can belong to a single team. This is shown by the foreign key TEAM_ID in the **REPORT** table.
7. **REPORT → PROJECTS**
    - Many reports can be related to a single project. This is shown by the foreign key P_ID in the **REPORT** table.

ER DIAGRAM


TEAM
TEAM_ID
TEAM_NAME
FORMS
MAKES
WORKS ON
User
U_ID
U_NAME
U_MAIL
TASKS
T_ID
T_NAME
DEADLINE
STATUS
DONE
GETS
LISTING
SPEND TIME
ATTENDS
NEED TIME
NOTIFY
NEED
TRACK_TIME
TRACK_ID
DATE
PROJECT
P_ID
P_NAME
START_DATE
END_DATE
CLIENT
C_ID
C_NAME
C_MAIL
MEETING
M_ID
M_TITLE
M_DATE_TIME
PROVIDES
HAS
HAS
SCHEDULES
S_DATE_TIME
S_DURATION
REPORTS
REPORT_ID
DATE
REMINDER
R_ID
R_TITLE
R_DATE_TIME
R_TYPE
MESSAGE

SCHEMA DIAGRAM



Query:

1) Retrieve all tasks that are dependent on meetings occurring before their deadline:

Showing rows 0 - 9 (10 total, Query took 0.0006 seconds.)

```
SELECT T.T_NAME AS Task_Name, M.M_TITLE AS Meeting_Title, T.T_DEADLINE AS Task_Deadline, M.M_DATE_TIME AS Meeting_Date FROM TASKS T JOIN MEETING M ON T.T_ID =
M.T_ID WHERE M.M_DATE_TIME < T.T_DEADLINE;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Task_Name	Meeting_Title	Task_Deadline	Meeting_Date
Task 1	Start	2024-01-15	2024-01-05
Task 2	Review	2024-02-15	2024-02-05
Task 3	Update	2024-03-15	2024-03-05
Task 4	Planning	2024-04-15	2024-04-05
Task 5	Explain	2024-05-15	2024-05-05
Task 6	Mid Way Met	2024-06-15	2024-06-05
Task 7	Brainstorming	2024-07-15	2024-07-05
Task 8	Report	2024-08-15	2024-08-05
Task 9	Demo	2024-09-15	2024-09-05
Task 10	Post Project Discussion	2024-10-15	2024-10-05

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2) Find all users who have worked on tasks in projects managed by teams other than their own:

Showing rows 0 - 1 (2 total, Query took 0.0008 seconds.)

```
SELECT U.U_NAME, T1.TEAM_NAME AS User_Team, P.P_NAME AS Project_Name, T2.TEAM_NAME AS Project_Team FROM USER U JOIN TEAM T1 ON U.U_ID = T1.U_ID JOIN TASKS
TA ON U.U_ID = TA.U_ID JOIN PROJECTS P ON TA.P_ID = P.P_ID JOIN TEAM T2 ON P.TEAM_ID = T2.TEAM_ID WHERE T1.TEAM_ID <> T2.TEAM_ID;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

U_NAME	User_Team	Project_Name	Project_Team
Shahtaz	Team Zed	Front End	Code Breaker
Shohana	Trio	Java Project	Team Zed

3) Find users who have reminders for tasks in the "Pending" status:

Showing rows 0 - 4 (5 total, Query took 0.0007 seconds.)

```
SELECT DISTINCT U.U_NAME as NAME , R.R_TITLE as TASK ,R_TYPE as REMINDER ,STATUS FROM REMINDER R JOIN TASKS T ON R.T_ID = T.T_ID JOIN USER U ON R.U_ID =
U.U_ID WHERE T.STATUS = 'Pending';
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

NAME	TASK	REMINDER	STATUS
Taposhi	Task 1	Email	Pending
Shahtaz	Task 4	Email	Pending
Tusar	Task 5	Notification	Pending
Tasnim	Task 7	Email	Pending
Shimanto	Task 10	Email	Pending

---

4) Retrieve clients, their associated projects, the team managing each project, the total number of tasks, and the percentage of completed tasks.

Showing rows 0 - 4 (5 total, Query took 0.0034 seconds.)

```
SELECT C.C_NAME AS Client_Name, C.C_MAIL AS Client_Email, T.TEAM_NAME AS Team_Name, COUNT(TA.T_ID) AS Total_Tasks, SUM(CASE WHEN TA.STATUS = 'Completed'
THEN 1 ELSE 0 END) AS Completed_Tasks FROM CLIENT C JOIN PROJECTS P ON C.C_ID = P.C_ID JOIN TEAM T ON P.TEAM_ID = T.TEAM_ID LEFT JOIN TASKS TA ON P.P_ID =
TA.P_ID GROUP BY C.C_NAME, C.C_MAIL, T.TEAM_NAME;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Client_Name	Client_Email	Team_Name	Total_Tasks	Completed_Tasks
Md. Shahidul Islam	shahid@gmail.com	Team Zed	2	0
Molla Rashied Hussein	rashied@gmail.com	Code Breaker	2	1
Molla Rashied Hussein	rashied@gmail.com	Trio	2	0
Shammi Akhtar	shamm@gmail.com	Trio	2	1
Tahera Parvin	tahera@example.com	Code Breaker	2	1

5)Get the total duration of meetings for each project:

Showing rows 0 - 4 (5 total, Query took 0.0005 seconds.)

```
SELECT P.P_NAME as PROJECT_NAME, SUM(S.S_DURATION) AS "Total_Duration(MIN)" FROM PROJECTS P JOIN TASKS T ON P.P_ID = T.P_ID JOIN MEETING M ON T.T_ID =
M.T_ID JOIN SCHEDULES S ON M.M_ID = S.M_ID GROUP BY P.P_NAME;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

PROJECT_NAME	Total_Duration(MIN)
Back End	150
DBMS Project	75
Front End	165
Java Project	90
UAP	105

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6)Completed Projects Query

Showing rows 0 - 4 (5 total, Query took 0.0010 seconds.)

```
SELECT P.P_ID AS Project_ID, P.P_NAME AS Project_Name, P.START_DATE AS Start_Date, P.END_DATE AS End_Date, T.TEAM_NAME AS Team_Name, C.C_NAME AS
Client_Name FROM PROJECTS P JOIN TEAM T ON P.TEAM_ID = T.TEAM_ID JOIN CLIENT C ON P.C_ID = C.C_ID WHERE P.END_DATE < CURRENT_DATE ORDER BY P.END_DATE ASC;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Project_ID	Project_Name	Start_Date	End_Date	Team_Name	Client_Name
1	UAP	2024-01-01	2024-02-01	Code Breaker	Molla Rashied Hussein
2	DBMS Project	2024-02-01	2024-03-01	Trio	Shammi Akhtar
3	Java Project	2024-03-01	2024-04-01	Team Zed	Md. Shahidul Islam
4	Front End	2024-04-01	2024-05-01	Code Breaker	Tahera Parvin
5	Back End	2024-05-01	2024-06-01	Trio	Molla Rashied Hussein

7) Best_User_Name

Showing rows 0 - 0 (1 total, Query took 0.0005 seconds.)

```
SELECT U_NAME as BEST_USER_ACCORDING_TO_WORK FROM USER WHERE U_ID = ( SELECT U_ID FROM TASKS WHERE STATUS = 'Completed' GROUP BY U_ID ORDER BY COUNT(T_ID)
DESC LIMIT 1 );
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

←T→	BEST_USER_ACCORDING_TO_WORK
☐ Edit Copy Delete	Partho

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8)Worst_User_Name

Showing rows 0 - 0 (1 total, Query took 0.0007 seconds.)

```
SELECT U.U_NAME AS WORST_USER_ACCORDING_TO_WORK FROM USER U WHERE U.U_ID = ( SELECT T.U_ID FROM TASKS T GROUP BY T.U_ID ORDER BY COUNT(CASE WHEN T.STATUS
!= 'Completed' THEN 1 END) DESC LIMIT 1 );
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

WORST_USER_ACCORDING_TO_WORK
Shimanto

---

9) All meetings scheduled for each team:

Showing rows 0 - 9 (10 total, Query took 0.0014 seconds.) [TEAM_NAME: **CODE BREAKER... - TRIO...**]

```
SELECT T.TEAM_NAME , M.M_TITLE AS MEETING_TITLE, M.M_DATE_TIME AS MEETING_DATE FROM MEETING M JOIN TEAM T ON M.TEAM_ID = T.TEAM_ID ORDER BY T.TEAM_NAME,
M.M_DATE_TIME;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

TEAM_NAME ▲ 1	MEETING_TITLE	MEETING_DATE
Code Breaker	Start	2024-01-05
Code Breaker	Planning	2024-04-05
Code Breaker	Brainstorming	2024-07-05
Code Breaker	Post Project Discussion	2024-10-05
Team Zed	Update	2024-03-05
Team Zed	Mid Way Met	2024-06-05
Team Zed	Demo	2024-09-05
Trio	Review	2024-02-05
Trio	Explain	2024-05-05
Trio	Report	2024-08-05

10)Tasks and Projects Assigned to Tusar, Shohana, and Shahtaz

Showing rows 0 - 2 (3 total, Query took 0.0004 seconds.)

```
SELECT U.U_NAME as NAME , T.T_NAME AS TASK_NAME, P.P_NAME AS PROJECT_NAME , T.STATUS AS STATUS, T.T_DEADLINE AS DEADLINE FROM USER U JOIN TASKS T ON
U.U_ID = T.U_ID JOIN PROJECTS P ON T.P_ID = P.P_ID WHERE U.U_NAME IN ('Tusar', 'Shohana', 'Shahtaz');
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

NAME	TASK_NAME	PROJECT_NAME	STATUS	DEADLINE
Shohana	Task 3	Java Project	In Progress	2024-03-15
Shahtaz	Task 4	Front End	Pending	2024-04-15
Tusar	Task 5	Back End	Pending	2024-05-15